Music Genre Classification using Machine Learning and Deep Learning

CS 4771 Final Project Report

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Abstract

This project implements a music genre classification system using both traditional machine learning and deep learning. Four models were trained on the GTZAN dataset: Logistic Regression, k-NN, Random Forest, and CNN. Surprisingly, Logistic Regression achieved the best test accuracy (74.0%), outperforming the CNN (66.4%), demonstrating that simpler models with well-engineered features can outperform complex deep learning on small datasets. A production-ready web application was developed for real-world deployment.

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1 Introduction & Problem Statement

1.1 Context and Motivation

Music genre classification is fundamental to modern music streaming services like Spotify and Apple Music, enabling personalized recommendations, intelligent playlists, and enhanced music discovery. This project develops an automated classification system for 10 music genres using machine learning.

1.2 Problem Definition

Given a 30-second audio clip, classify it into one of ten genres: Blues, Classical, Country, Disco, Hip-Hop, Jazz, Metal, Pop, Reggae, or Rock. The goal is to compare traditional machine learning approaches with deep learning methods.

1.3 Dataset: GTZAN Genre Collection

- 1,000 audio tracks (100 per genre)
- 30 seconds each, WAV format
- 22,050 Hz sampling rate, mono, 16-bit
- Perfectly balanced dataset
- Split: 70% training, 15% validation, 15% test

1.4 Project Scope

- Extract handcrafted audio features (MFCCs, chroma, spectral, temporal)
- Train baseline models: Logistic Regression, k-NN, Random Forest
- Develop CNN for end-to-end learning from mel-spectrograms
- Build production-ready Flask web application
- Cross-platform support (Windows 11 and Ubuntu Server)

2 Methodology / Approach

2.1 Technology Stack

Python 3.9+, PyTorch 2.0+, scikit-learn, librosa (audio processing), Flask (web framework), NumPy/Pandas, HDF5 storage.

2.2 Data Preprocessing

Audio files were loaded at 22,050 Hz and normalized. For CNN training, each 30-second track was divided into overlapping 3-second segments (50% overlap), creating approximately 10 segments per track for data augmentation.

2.3 Feature Extraction

Handcrafted Features (58 dimensions):

- **MFCCs**: 13 coefficients capturing timbral texture
- **Chroma**: 12 pitch class features for harmonic content
- **Spectral**: Centroid, bandwidth, rolloff, contrast
- **Temporal**: Zero-crossing rate, RMS energy, tempo

Mel-Spectrograms for CNN:

- 128 mel bands × 130 time frames (3-second segments)
- Hop length: 512, FFT size: 2048, Hann window
- Log-scale transformation for perceptual representation

2.4 Models Implemented

1. Logistic Regression: L2 regularization (C=1.0), L-BFGS solver, multinomial strategy

2. k-Nearest Neighbors: k=5, cosine distance metric, ball tree algorithm
3. Random Forest: 100 trees, unlimited depth, bootstrap sampling
4. CNN Architecture:

- Input: 1×128×130 mel-spectrogram
- Three conv blocks (32→64→128 filters, 3×3 kernels)
- Batch normalization and dropout (25%) after each block
- MaxPooling (2×2) for dimension reduction
- Global Average Pooling
- Dense layers: 128 neurons → 10 classes (softmax)
- Adam optimizer (lr=0.001), batch size=32
- Early stopping (patience=10 epochs)

2.5 Web Application

Flask REST API with endpoints for prediction, model listing, and health checks. Modern responsive frontend with drag-and-drop upload, real-time predictions, and confidence visualization. Deployed with Gunicorn on Ubuntu Server.

2.6 Changes from Proposal

Added k-NN baseline, expanded feature set with spectral contrast, implemented overlapping segmentation strategy, developed production web application beyond CLI tool, and added cross-platform support.

3 Results & Analysis

3.1 Overall Performance

Table 1: Model Performance Comparison (Test Set: 300 samples)

Model	Accuracy	Precision	Recall	F1
Logistic Regression	**74.0%**	**0.741**	**0.740**	**0.739**
k-NN	70.0%	0.703	0.700	0.690
Random Forest	68.3%	0.670	0.683	0.672
CNN (segment-level)	66.4%	0.730	0.651	0.622

Key Finding: Logistic Regression unexpectedly outperformed all models, including the CNN. This demonstrates that simple linear classifiers with well-engineered features can be superior to deep learning on small datasets (1,000 samples insufficient for CNN).

3.2 Per-Genre Performance

Table 2: Logistic Regression Per-Genre Results (Best Model)

Genre	**Precision**	**Recall**	**F1-Score**
Classical	0.903	0.933	0.918
Metal	0.824	0.933	0.875
Jazz	0.889	0.800	0.842
Country	0.788	0.867	0.825
Blues	0.846	0.733	0.786
Pop	0.727	0.800	0.762
Disco	0.692	0.600	0.643
Reggae	0.643	0.600	0.621
Hip-Hop	0.548	0.567	0.557
Rock	0.548	0.567	0.557

Best Genres: Classical (F1=0.918) and Metal (F1=0.875) due to distinctive instrumentation and spectral characteristics.

Challenging Genres: Hip-Hop and Rock (F1=0.557) due to high intra-genre variability and overlap with other genres.

3.3 CNN Analysis

Table 3: CNN Per-Genre Performance (1,497 segments)

Genre	**Precision**	**Recall**	**F1-Score**
Classical	0.918	0.967	0.942
Metal	0.879	0.827	0.852
Jazz	0.616	0.946	0.746
Pop	0.623	0.880	0.729
Reggae	0.610	0.813	0.697
Country	0.557	0.685	0.614
Disco	0.873	0.367	0.516
Rock	0.389	0.640	0.484
Blues	0.872	0.227	0.360
Hip-Hop	0.960	0.161	0.276

Critical Issue: CNN severely under-predicted Hip-Hop (16.1% recall) and Blues (22.7% recall), indicating class imbalance problems despite high precision.

3.4 Why Logistic Regression Won

1. **Well-Engineered Features**: 58 handcrafted features captured essential genre characteristics designed by domain experts
2. **Appropriate Complexity**: Simple linear model avoided overfitting with limited data (700 training samples)
3. **Linear Separability**: Genres more linearly separable in feature space than anticipated
4. **Robust to Small Data**: Linear models generalize well with limited samples; L2 regularization prevented overfitting

3.5 Why CNN Underperformed

1. **Insufficient Data**: 700 tracks (7,000 segments) too few for deep learning; CNNs typically need 10,000+ samples
2. **Severe Class Imbalance**: Hip-Hop recall 16.1% and Blues 22.7% indicate model learned to ignore these classes
3. **No Data Augmentation**: Missing pitch shifting, time stretching, SpecAugment
4. **Architecture Limitations**: Simple 3-block CNN without residual connections or transfer learning
5. **Segment vs. Track Evaluation**: CNN evaluated on 3-second segments without track-level majority voting

3.6 Random Forest Disappointment

Random Forest (68.3%) placed third, with catastrophic Rock genre failure (F1=0.275, only 23.3% recall). This suggests overfitting to training data and that ensemble complexity didn't help with linearly separable features.

3.7 Confusion Patterns

Logistic Regression Confusions:

- Reggae $\rightarrow$ Hip-Hop (30%): Both share bass-heavy, rhythmic characteristics
- Rock Disco (16.7% each way): Era-based rhythmic similarities
- Classical minimal confusion (only 2 errors): Well-separated in feature space

3.8 Comparison Against Proposal Objectives

Original Proposal Goals vs. Actual Results:

Table 4: Proposal Objectives Achievement Assessment

Proposal Objective	**Status**	**Result**
Train baseline models (LR, k-NN, RF)	Met	All three trained
Achieve 60-70% baseline accuracy	Exceeded	LR: 74.0%
Implement CNN on spectrograms	Met	CNN implemented
CNN outperforms baselines	Not Met	CNN: 66.4% ¡ LR: 74.0%
Evaluate with accuracy, precision, recall, F1	Met	All metrics reported
Deliver working classifier	Exceeded	CLI + Web app

Justification for Unmet Objective (CNN Performance):

The proposal anticipated the CNN would achieve the highest accuracy based on literature showing deep learning typically outperforms traditional ML on audio tasks (80-90% accuracy). However, our CNN achieved only 66.4% test accuracy, underperforming all baseline models. This discrepancy is justified by:

1. **Dataset Size Mismatch**: Literature results use 10,000+ samples; GTZAN has only 1,000, insufficient for deep learning
2. **No Data Augmentation**: Published CNN results employ extensive augmentation (pitch shift, time stretch, SpecAugment); we used only segmentation
3. **Class Imbalance**: CNN severely under-predicted Hip-Hop (16.1% recall) and Blues (22.7% recall), not addressed during training
4. **Evaluation Method**: CNN evaluated on segments vs. baselines on full tracks; lacking track-level majority voting
5. **No Transfer Learning**: Literature often uses pre-trained models; we trained from scratch

Exceeded Expectations:

- Baseline accuracy exceeded target (74% vs. 60-70% goal)
- Delivered production web application beyond proposed CLI tool
- Implemented cross-platform support not in original proposal

- Added fourth baseline model (k-NN) for comprehensive comparison

Educational Value of Unmet Objective:

The CNN underperformance, while unexpected, provides critical insight: *deep learning is not always superior, especially with small datasets*. This negative result validates fundamental ML principles about model complexity matching data size, making it educationally more valuable than simply reporting expected success.

3.9 Comparison to Literature

Published GTZAN results: 60-70% (simple baselines), 70-80% (feature-engineered ML), 80-92% (deep learning).

Our results: Logistic Regression 74% places solidly in feature-engineered ML range. CNN 66.4% below typical CNN performance (80-90%), confirming insufficient data and need for augmentation/transfer learning.

3.10 Key Lessons

1. **Model Complexity Must Match Data Size**: Simpler models outperform complex ones with limited data

2. **Feature Engineering Valuable**: Domain knowledge encoding outperformed learned representations with small datasets

3. **Class Imbalance Critical**: Must implement class weights, focal loss, or SMOTE

4. **Data Augmentation Essential**: Deep learning requires augmentation with small datasets

5. **Evaluation Strategy Matters**: Segment-level vs. track-level evaluation affects comparison fairness

4 Conclusion

4.1 Summary

This project successfully implemented a complete music genre classification pipeline with four models, comprehensive evaluation, and production deployment. Despite expectations, Logistic Regression (74.0%) outperformed the CNN (66.4%), providing valuable lessons about model selection for small datasets.

Major Accomplishments:

- Four working models with thorough evaluation

- 58-dimensional handcrafted feature extraction
- CNN with mel-spectrogram processing
- Production web application with REST API
- Cross-platform deployment (Windows + Linux)

4.2 Critical Findings

1. Simple models with good features outperform complex models with insufficient data
2. 1,000 samples insufficient for deep learning without heavy augmentation
3. Class imbalance severely impacts CNN performance (Hip-Hop 16.1% recall)
4. Classical music universally easy (>84% F1 all models); Rock universally hard
5. Genre boundaries are inherently fuzzy and subjective

4.3 Recommendations for Improvement

For CNN:

- Implement class weights/focal loss for imbalance
- Add data augmentation (pitch shift, time stretch, SpecAugment, mixup)
- Use residual connections and attention mechanisms
- Apply transfer learning from pre-trained audio models
- Implement proper track-level majority voting

For System:

- Ensemble Logistic Regression + CNN predictions
- Train on larger datasets (FMA, Million Song Dataset)
- Multi-label classification for genre overlap
- Hierarchical classification for subgenres

4.4 Final Remarks

While the CNN didn't achieve expected performance, this outcome provides valuable educational insight: machine learning requires careful consideration of data size, model complexity, and evaluation strategy. The project successfully demonstrates complete ML pipeline development, rigorous comparison methodology, critical analysis of failure modes, and production-ready deployment.

The unexpected result—simpler models winning—reinforces fundamental ML principles: *more complex is not always better*. With 1,000 samples, well-engineered features and linear models proved superior to deep learning, validating Occam's Razor in machine learning practice.

Most importantly, the project delivers practical value through an accessible web interface, comprehensive documentation, and extensible architecture, bridging academic machine learning concepts with real-world software engineering.

5 References

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